

(1001CJA102120092)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

25-04-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C,D	A,C,D	B,D	A,B,C,D	A,B	A,B	A,B,C	B	A	D
	Q.	11	12	13							
	A.	C	B	A							
SECTION-II	Q.	1	2	3	4	5					
	A.	3.50	1.50	1.60	1.00	3.24					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,D	A,C	A,C,D	C,D	A,B,D	B,C	A,C	B	C	B
	Q.	11	12	13							
	A.	B	D	A							
SECTION-II	Q.	1	2	3	4	5					
	A.	8.00	29.00	63.50	12.00	12.50					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,B	B,D	B,C	A,D	A,D	B,C	A	B	D
	Q.	11	12	13							
	A.	B	B	D							
SECTION-II	Q.	1	2	3	4	5					
	A.	4.00	0.25	137.00	404.00	4.00					

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25-04-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	C	C	B	C	C	A,B	A,B	A,B,C,D
	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,C	A,B	B,C,D	A,C	A	A	C	C		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	D	B	D	D	C	D	B,C	A,B,D	A,C,D
	Q.	11	12	13	14	15	16	17	18		
	A.	C,D	A,C	A,C	A,B,C	C	B	C	D		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	C	B	C	D	A,C,D	A,B,C	B,C,D
	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,D	B,C,D	B,C,D	A,B	D	A	C	D		

PAPER-1**PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A, B, C, D)****Sol.** (A) Photo-electric effect is instantaneous phenomenon.

(B) Bright light has more intensity as compared to dim light of same frequency.

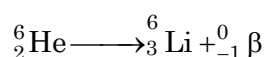
Intensity $\propto$ photon-flux

$$\begin{aligned} E &= \phi + k_{\max} \\ \downarrow &\quad \downarrow \\ \text{constant} &\quad \text{constant} \end{aligned}$$

$\therefore k_{\max}$ is constant

(C) $\therefore E = h\nu$

(D) Below threshold frequency photo-electric effect doesn't occur.

2. **Ans. (A, C, D)****Sol.** B.E. per nucleon = $\frac{\text{B.E.}}{\text{total no. of nucleon}}$ $\therefore$ mass of neutron and proton are different hence their rest mass energy,3. **Ans. (B, D)****Sol.** From lens maker formula,

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\text{Now, } \frac{1}{f_1} = \left(\frac{3}{2} - 1 \right) \left(\frac{1}{R} - \frac{1}{-R} \right) = \frac{1}{R}$$

$$\& \frac{1}{f_2} = \left(\frac{4}{3} - 1 \right) \left(\frac{1}{\infty} - \frac{1}{R} \right) = -\frac{1}{3R}$$

$$\therefore \frac{1}{f_{\text{eq}}} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{R} - \frac{1}{3R} = \frac{2}{3R}$$

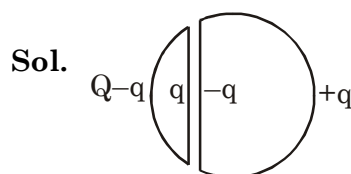
$$f_{\text{eq}} = \frac{3R}{2}$$

Now, image coincides with the object when ray of light retraces its path i.e. fall normally on plane mirror. This is possible only when object is at focus of lens system.

$$\therefore f_{\text{eq}} = 15 \text{ cm} = \frac{3R}{2}$$

$$\therefore R = 10 \text{ cm}$$

$$\text{Also } f_1 = R = 10 \text{ cm}$$

4. **Ans. (A, B, C, D)**

(A) Due to negligible separation between two parts, capacitance is very high

$$\therefore C = \frac{\epsilon_0 A}{d}$$

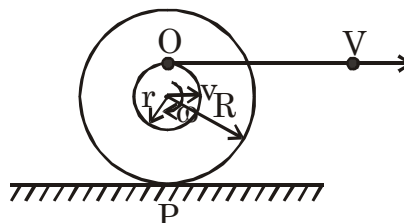
Hence energy stored will be negligible

$$\therefore \text{Energy} = \frac{q^2}{2C}$$

(B, C) In equilibrium, system will attain minimum electrostatic energy. So charge Q will distribute uniformly on sphere to have minimum energy.

(D) Due to uniform charge distribution through out the surface of sphere,

$$E(r > R) = \frac{Q}{4\pi\epsilon_0 r^2}$$

5. **Ans. (A, B)****Sol.** $\therefore$ No slipping at point of contact P on Q

$$\therefore v = \omega R \quad \dots(1) \text{ (for P)}$$

$$V = v + \omega r \quad \dots(2) \text{ (for Q)}$$

from equation (1) and (2)

$$V = \omega R + \omega r \Rightarrow \omega = \frac{V}{R + r}$$

$$\text{and } v = \frac{VR}{R + r} \text{ (from (i))}$$

6. Ans. (A, B)

Sol. $\rho h = \frac{1}{2} \rho v^2$

$\Rightarrow v = \sqrt{2gH}$

$Q = \text{Area} \times v = \ell h \sqrt{2gh}$

7. Ans. (A, B, C)

Sol. Power transmission at node or antinode is zero

Elements in between node and adjoining antinode do exchange energy.

8. Ans. (B)

9. Ans. (A)

10. Ans. (D)

Sol. In case (i) $m \frac{dv}{dt} = -\frac{(B\ell v)}{R} \ell B + mg$

so conductor will acquire terminal velocity

In case (ii) $q = CB\ell v$ and $m \frac{dv}{dt} = mg - I\ell B$

$I = \frac{dq}{dt}$

So acceleration of conductor will be constant

In case (iii) $B\ell v = -L \frac{dI}{dt}$

So conductor will perform SHM

In case (iv) conductor may be stationary.

Sol. (11 to 13)

$\therefore I_R = 4I_0 \cos^2\left(\frac{\Delta\phi}{2}\right)$ and $\Delta\phi = \frac{2\pi}{\lambda}(d \sin \theta)$

11. Ans. (C)

If $d = 2\lambda$ and $\sin \theta = \frac{1}{4}$

$\Delta\phi = \frac{2\pi}{\lambda}\left(2\lambda \times \frac{1}{4}\right) = \pi$

$I_R = 4I_0 \cos^2\left(\frac{\pi}{2}\right) = 0$

Hence (C)

12. Ans. (B)

If $d = \frac{3\lambda}{2}$ and $Q = \frac{\pi}{6}$

$\Delta\phi = \frac{2\pi}{\lambda}\left(\frac{3\lambda}{2} \times \frac{1}{2}\right) = \frac{3\pi}{2}$

$\therefore I_R = 4I_0 \cos^2\left(\frac{3\pi}{4}\right) = 4I_0 \times \frac{1}{2} = 2I_0$

13. Ans. (A)

If $d = \lambda$, $\sin \theta = \frac{1}{6}$

$\Delta\phi = \frac{2\pi}{\lambda}\left(\lambda \times \frac{1}{6}\right) = \frac{\pi}{3}$

$\therefore I_R = 4I_0 \cos^2\left(\frac{\pi}{6}\right) = 4I_0 \times \frac{3}{4} = 3I_0$

SECTION-II

1. Ans. 3.50

Sol. $P = k g^\alpha L^\beta \rho_{\text{hel}}^\gamma \rho_{\text{air}}^\delta \Rightarrow \gamma + \delta = 1$

$\alpha + \beta - 3(\gamma + \delta) = 2 \Rightarrow \beta = \frac{7}{2}$

2. Ans. 1.50

3. Ans. 1.60

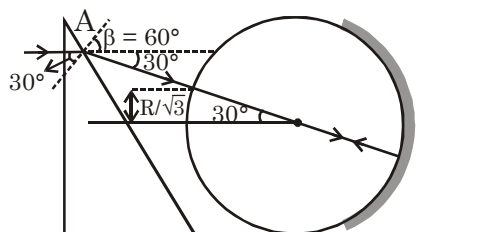
Sol. $\tau = -C\theta$, $\alpha = -\frac{C}{I}\theta$

$\therefore \omega = \sqrt{\frac{C}{I}} \Rightarrow C = I\omega^2$

$C = I\left(\frac{2\pi}{T}\right)^2 = \frac{I 4\pi^2}{T} = \frac{MR^2}{2} \times 4\pi^2$

$= \frac{2 \times 10^{-3} \times 4 \times 10^{-4}}{2} \times 10 \times 4 = 1.60 \times 10^{-5}$

4. Ans. 1.00



Sol.

After refraction from prism the ray is incident normally on the sphere & hence after reflection from polished surface they will retrace their path.

5. Ans. 3.24

Sol. For PO box (wheat stone bridge) under balanced condition

$\frac{P}{Q} = \frac{R}{S}$ where S is Unknown resistance.

$S = \frac{Q}{P} \times R$ [Here $\frac{P}{Q} = 100$ given]

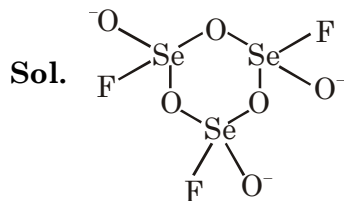
From given graph galvanometer shows zero deflection at $R = 323.75 = 324\Omega$ [Upto 3 SD]

So, $S = 3.24 \Omega$

PART-2 : CHEMISTRY**SOLUTION****SECTION-I**

1. Ans. (A,D)

2. Ans. (A,C)



Sol. • Around each Se: $\left. \begin{array}{l} \text{no. of LP} = 1 \\ \text{no. of BP} = 1 \end{array} \right\} \Rightarrow \text{Total } 5e^- \text{ pairs}$

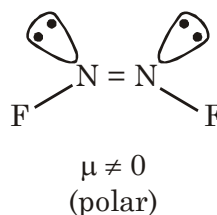
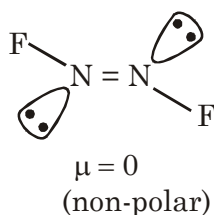
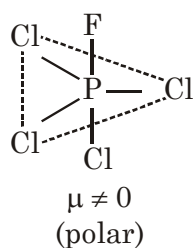
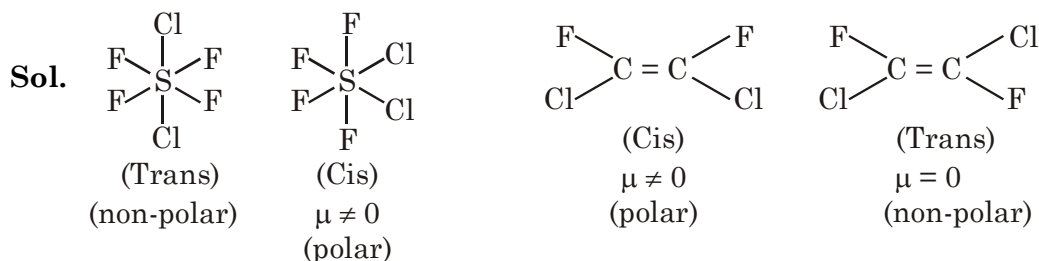
- Cycle is non-planar, so it can't be aromatic.
- About Se, 1 LP and 4 BP $\Rightarrow$ see-saw shape
- Due to presence of lone pair on bridged O, geometry about it, is non-linear.

3. Ans. (A,C,D)

4. Ans. (C,D)

Sol. (C) $\rightarrow \text{FeCl}_3 + 3\text{NaOH} \rightarrow \text{Fe(OH)}_3 + 3\text{NaCl}$ $\text{NaOH} \rightarrow \text{Na}^+ + \text{OH}^-$ Common ion i.e. OH^- adsorb preferentially.

5. Ans. (A,B,D)



6. Ans. (B,C)

7. Ans. (A,C)

8. Ans. (B)

9. Ans. (C)

10. Ans. (B)

Sol. For question number (8, 9, 10) Theory based

11. Ans. (B)

12. Ans. (D)

13. Ans. (A)

Sol:

	ionisation	linkage	Geometrical	optical	no. of stereoisomers
$[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$	✓	✓	✗	✗	0
$[\text{Cr}(\text{NH}_3)_4\text{Cl}_2] \text{Br}$	✓	✗	✓	✗	2
$[\text{Cr}(\text{en})_2\text{Cl}_2]\text{Cl}$	✗	✗	✓	✓	3
$[\text{Co}(\text{gly})_3]$	✗	✗	✓	✓	4

SECTION-II

1. Ans. 8.00

Sol. $\Lambda_m = \frac{1000 \times 3.4 \times 10^{-5}}{1.25 \times 10^{-3}} = 27.2$

$$\Lambda_m^\infty = (90 + 380 - 130) = 340$$

$$\alpha = \frac{27.2}{340} = 0.08$$

$$\alpha = 8 \times 10^{-2}$$

$$\therefore \boxed{y = 8}$$

2. Ans. 29.00

Sol. $3s = 0$

$$3p = 1 \times 3 = 3$$

$$3d = 2 \times 5 = 10$$

$$4s = 0$$

$$4p = 1 \times 3 = 3$$

$$4d = 2 \times 5 = 10$$

$$5s = 0$$

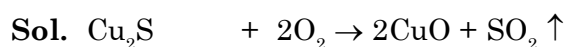
$$5p = 1 \times 3 = 3$$

$$6s = 0$$

$$\text{Angular node (l)} = 0 + 3 + 10 + 0 + 3 + 10 + 0 + 3$$

$$l = 29$$

3. Ans. 63.50



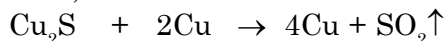
$$39.75 \text{ (g)} \quad (\text{excess})$$

↓

$$0.25 \text{ mol}$$

$$\text{Hence, moles of CuO formed} = 0.5 \text{ mol}$$

Now,



$$(\text{excess}) \quad 0.5 \text{ mol}$$

$$\text{Moles of Cu formed} = 1 \text{ mol}$$

$$\text{Mass of elemental Cu} = 63.50 \text{ g}$$

4. Ans. 12.00

Sol. $x = 3, y = 3$

5. Ans. 12.50

Sol. $x = 3$ because reactions (i), (ii) & (iii) are correct.

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A,C)**

Sol. $I.F = e^{\int \frac{2\cos x - x \sin x}{x \cos x} dx} = e^{2\ln x - \ln \sec x} = x^2 \cos x$

$\therefore$ solution

$$y(x^2 \cos x) = \int x^4 dx + C$$

$$y(x^2 \cos x) = \frac{x^5}{5} + C$$

$$y = \frac{x^3}{5 \cos x} + \frac{C}{x^2 \cos x}$$

$$y(1) = \left(\frac{1}{5} + C \right) \frac{1}{\cos 1} = \frac{1}{\cos 1} \Rightarrow C = \frac{4}{5}$$

$$y = \frac{x^5 + 4}{x^2 \cdot 5 \cos x} = \frac{x^3}{5 \cos x} + \frac{4}{5x^2 \cos x}$$

$$y'(1) = \frac{1}{5} \left[\sec 1 \cdot \tan 1 + \frac{3}{\cos 1} - \frac{8}{\cos 1} + 4 \sec 1 \tan 1 \right]$$

$$= \frac{1}{5} \left[5 \frac{\tan 1}{\cos 1} - \frac{5}{\cos 1} \right] = \frac{(\tan 1 - 1)}{\cos 1}$$

$$\lim_{x \rightarrow \infty} x^\alpha y \cos x = \lim_{x \rightarrow \infty} \left(\frac{x^{3+\alpha}}{5} + \frac{4}{5} x^{\alpha-2} \right)$$

$$= 0 + 0 = 0 \text{ [if } \alpha < -3]$$

2. **Ans. (A,B)**

Sol. $\vec{p} + \vec{q} + \vec{r} = 0$

So, $\vec{p}, \vec{q}, \vec{r}$ can form a triangle

$$\vec{p} \cdot \vec{c} = 0$$

So $\vec{p} \perp \vec{c}$; and $\vec{q} \perp \vec{a}$ and $\vec{r} \perp \vec{b}$

option (C) rejected for $\vec{a} = \vec{b} = \vec{c} = \hat{i}$

3. **Ans. (B,D)**

Sol. $I = \int_0^{\frac{\pi}{2}} \log(\sin x) dx$

$$x = 3t$$

$$I = 3 \int_0^{\frac{\pi}{6}} \log(\sin 3t) dt$$

$$= 3 \left[\int_0^{\frac{\pi}{6}} \log(\sin t) dt + \int_0^{\frac{\pi}{6}} \log(4 \cos^2 t - 1) dt \right]$$

$$-\frac{\pi}{2} \ln 2 = 3 \left[\int_0^{\frac{\pi}{6}} \log(\sin t) dt + \int_0^{\frac{\pi}{6}} \log(4 \cos^2 t - 1) dt \right]$$

$$\int_0^{\frac{\pi}{6}} \ln(\sin t) dt + \int_0^{\frac{\pi}{6}} \ln(4 \cos^2 t - 1) dt = -\frac{\pi}{6} \ln 2$$

4. **Ans. (B,C)**

Sol. $f'(x) = e^{x^4 - x^2} \cdot 2x$

5. **Ans. (A,D)**

Sol. $f(x) = f'(x) \cdot f''(x)$

$$f'(x) = (f''(x))^2 + f'(x) \cdot f'''(x)$$

$$\frac{f(x)}{f''(x)} = f''(x)^2 + \frac{f(x)}{f''(x)} \cdot f'''(x)$$

$$f = (f'')^3 + f \cdot f'''$$

6. **Ans. (A,D)**

Sol. $b = 0$

$$\frac{a}{3} - \frac{14}{c} = 3$$

$$\frac{a}{3} - 3 = \frac{14}{c} \Rightarrow \frac{a-9}{3} = \frac{14}{c}$$

$$a-9 = \frac{42}{c} \Rightarrow a = 9 + \frac{42}{c}$$

$$\therefore c = 1, 2, 3, 6$$

$$a = 51, 30, 23, 16$$

7. **Ans. (B,C)**

Sol. $x^2 = \frac{10^{2020} - 1}{9} - \frac{2(10^{1010} - 1)}{9}$

$$= \frac{(10^{1010})^2 - 2 \cdot 10^{1010} + 1}{9} = \left(\frac{10^{1010} - 1}{3} \right)^2$$

$$x = \frac{10^{1010} - 1}{3}$$

$$a = 1010 = 2.505 = 2.5.101$$

$$b = 3$$

8. **Ans. (A)**

9. **Ans. (B)**

10. **Ans. (D)**

Sol. Centroid = $\left(\frac{2\alpha - 4a}{3}, 0 \right)$

$$\text{Circumcenter} = \left(\frac{\alpha + 2a}{2}, \frac{\beta}{4} \right)$$

$$\text{Orthocenter} = \left(\alpha - 6a, -\frac{\beta}{2} \right)$$

11. Ans. (B)

12. Ans. (B)

13. Ans. (D)

Sol. All 7 digits possible numbers = 9×10^6

(i) If $a_1 > a_2 > a_3 \rightarrow a_7 = {}^{10}C_7 = 120$

(ii) If order remains same on reversing
then = 9×10^3 (First & last place has 9 options. 2nd & 6th = 10 options
3rd & 5th = 10 options
4th = 10 options)

(iii) If even and odd are consecutive & starts with odd = 5^5

(iv) If $a_1 < a_2 < a_3 \dots \dots \dots < a_7 = {}^9C_7 = 36$

All digit possible numbers = 9×10^4

(i) If $a_1 > a_2 > a_3 > a_4 > a_5 = {}^{10}C_5 = 252$

(ii) If order remains same on reversing
then = 9×10^2 (First & last place has 9 options. 2nd & 4th = 10 options
3rd place = 10 options)

(iii) If even and odd are consecutive & starts with odd = 5^5

(iv) If $a_1 < a_2 < a_3 < a_4 < a_5 = {}^9C_5 = 126$

SECTION-II

1. Ans. 4

Sol. $\int \left(1 + x \left(\cos x - \frac{\cos x}{\sin^2 x} \right) \right) e^{\sin x + \csc x} dx$
 $\int e^{\sin x + \csc x} dx + \int x \left(\cos x - \frac{\cos x}{\sin^2 x} \right) e^{\sin x + \csc x} dx$
 $= x e^{\sin x + \csc x} + C$

$$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} e^2; f\left(-\frac{\pi}{2}\right) = \frac{-\pi}{2} e^{-1-1} + C \Rightarrow C = 0$$

$$\Rightarrow a = 2, b = 2$$

$$\therefore a + b = 4$$

2. Ans. 0.25

Sol. Let $g(x) = f^{-1}(x)$

$$\Rightarrow f(g(x)) = x \Rightarrow g'(x) = \frac{1}{f'(g(x))}$$

$$g'(-7) = \frac{1}{f'(g(-7))}$$

$$f(0) = 1 - 8 = -7$$

$$\therefore g(-7) = 0$$

$$\therefore g'(-7) = \frac{1}{f'(0)} = \frac{1}{4-16+16} = 0.25$$

3. Ans. 137

Sol. www + Bwww + wBwww + wwBww

$$\frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} + \frac{2}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} + \frac{3}{5} \times \frac{3}{5} \times \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} + \frac{3}{5} \times \frac{2}{5} \times \frac{4}{5} \times \frac{2}{5} \times \frac{1}{5}$$

$$= \frac{6 \times 25 + 48 + 54 + 48}{625 \times 5} = \frac{12}{125}$$

4. Ans. 404

$$\text{Sol. } x \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{2020 \times 1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{2020 \times 1}$$

$$\Rightarrow x^{2020} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{2020 \times 1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{2020 \times 1}$$

$$\sum_{i=1}^{2020} R_i^2 = 1 + 1 + \dots + 1 = 2020$$

$$\frac{2020}{5} = 404$$

5. Ans. 4

Sol. $2y + 4(y - a)^2 = 4$

$$\Rightarrow y + 2y^2 + 2a^2 - 4ya = 2$$

$$\Rightarrow 2y^2 + (1 - 4a)y + 2(a^2 - 1) = 0 \quad \dots\dots(i)$$

$$\Delta \geq 0 \quad (1 - 4a)^2 - 16(a^2 - 1) \geq 0$$

$$1 + 16a^2 - 8a - 16a^2 + 16 \geq 0 \Rightarrow a \leq 17/8$$

If both roots of (i) are positive

$$\therefore 1 - 4a \leq 0; a \geq 1/4$$

$$\& 2(2a^2 - 2) > 0 \Rightarrow a > 1 \text{ or } a < -1$$

$$\therefore a \in \left(1, \frac{17}{8} \right) \quad \dots\dots(ii)$$

If roots have opp. sign $-1 \leq a \leq 1 \quad \dots\dots(iii)$

$$\therefore a \in \left[-1, \frac{17}{8} \right] \therefore a_{\text{int}} = -1, 0, 1, 2$$

$$\therefore a_{\text{int}} = -1, 0, 1, 2$$

PAPER-2**PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (B)**

Sol. $\frac{u}{V} = \frac{r}{R}$
 $R^2 = r^2 + d^2$
 $\Rightarrow d = R\sqrt{1 - \frac{u^2}{V^2}}$

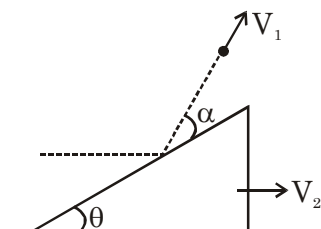
2. **Ans. (A)**

Sol. Let $\frac{m_2}{m_1} = q$

conserving horizontal momentum

$$u = V_1 \cos(\alpha + \theta) + qV_2$$

conserving kinetic energy



$$u^2 = V_1^2 + q^2 V_2^2$$

along surface of wedge for particle

$$u \cos \theta = V_1 \cos \alpha$$

$$\text{for second impact } V_1 \cos(\alpha + \theta) = V_2$$

$$q = \frac{\tan^2 \theta}{1 - \tan^2 \theta}$$

$$\text{Now } \frac{m_1}{m_2} = \cot^2 \theta - 1$$

3. **Ans. (C)**

Sol. $\frac{F}{\pi r_1^2} = \frac{4F}{\pi r_2^2} \Rightarrow \frac{r_2}{r_1} = 2$

4. **Ans. (C)**

Sol. $\Delta V = \frac{2kq}{r}$

$$\text{and } 2Ex = \frac{2kq}{r} \quad (x = l/2)$$

$$\therefore q = \frac{Er x}{k}$$

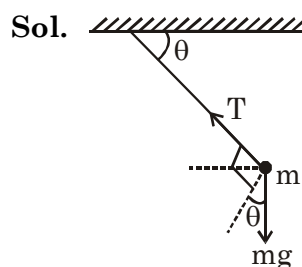
$$F = \frac{kq^2}{x^2} = \frac{E^2 r^2}{k} = \text{constant}$$

5. **Ans. (B)**

Sol. $dB = \frac{\mu_0 \left(\frac{\lambda dx}{2\pi/w} \right)}{2x} = \frac{\mu_0 \lambda w}{4\pi} \frac{dx}{x}$

$$B = \frac{\mu_0 \lambda w}{4\pi} \int_{l/2}^l \frac{dx}{x}$$

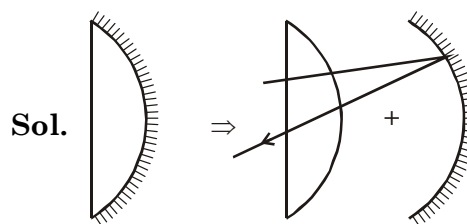
$$= \frac{\mu_0 \lambda w}{4\pi} \log_e 2$$

6. **Ans. (C)**

$$a_t = g \cos \theta$$

$$(a_t)_y = a_t \cos \theta$$

$$= g \cos^2 \theta$$

7. **Ans. (C)**

$$P = 2P_L + P_m$$

$$= 2 \times \frac{(\mu - 1)}{R} - \frac{1}{-R/2} = \frac{2\mu - 2}{R} + \frac{2}{R}$$

$$= \frac{2\mu}{R} = \frac{3}{R}$$

$$\Rightarrow P = 2P_L + P_m = 2 \times \left(\frac{\mu - 1}{R} \right) = \frac{1}{R}$$

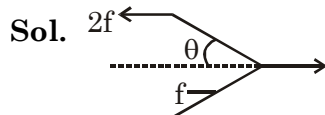
8. Ans. (A,B)

Sol. $\eta_1 = \frac{W}{Q_{1-2} + Q_{2-3}}$ and $\eta_2 = \frac{W}{Q_{1-3}}$

Also $Q_{1-2} + Q_{2-3} + Q_{3-1} = W$

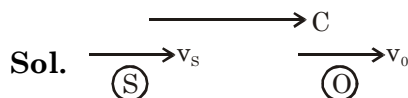
Also $Q_{3-1} = -Q_{1-3}$

9. Ans. (A,B)



$\sin \theta = \frac{a/2}{l}$ and $\cos \theta = \frac{a}{l/2}$

10. Ans. (A,B,C,D)



velocity of sound in frame of observer

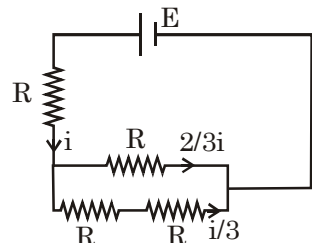
$\Rightarrow (C - v_o) < C$

$f_r = \left(\frac{C - v_o}{C - v_s} \right) f_0$

Recorded frequency doesn't depends on separation between source & Observer

11. Ans. (A,B,C)

Sol. As given all bulb are identical
 $\Rightarrow$ all having same resistance



$R_{eq} = \frac{2R}{3} + R = \frac{5R}{3}$

$i = \frac{E}{5R} \times 3$

$B_1 = i^2 R$ $B_3 = B_4 = \left(\frac{i}{3} \right)^2 R$

$B_2 = \left(\frac{2i}{3} \right)^2 R$

12. Ans. (A,B)

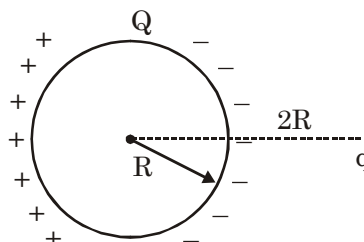
Sol. $k = \frac{\Delta \theta}{\Delta t} \times \frac{1}{4\pi r^2} \frac{dr}{dT} = P \frac{r^3}{R^3}$

$\therefore dT = \frac{P}{4\pi k r^3} r dr$

$\Rightarrow \Delta T = \frac{P}{8\pi k R}$

13. Ans. (B,C,D)

Sol. Due to induction charge distribution on the surface of sphere is non uniform.



$v_c = \frac{kQ}{R} + \frac{k(q)}{3R} + v_{induced}$ ($v_{induced}$ at centre = 0)

$v_c = \frac{1}{4\pi \epsilon_0} \times \frac{Q}{R} + \frac{1}{4\pi \epsilon_0} \times \frac{q}{3R}$

$v_{x=\frac{R}{2}} \Rightarrow v_{at \text{ centre}}$

Because inside conductor

net $\vec{E} = 0 \Rightarrow$ potential remain constant

14. Ans. (A,C)

Sol. $V_1 = \frac{6R_1}{R_1 + R_2}$ and $V_2 = \frac{6R_2}{R_1 + R_2}$

$\Rightarrow \frac{R_2}{R_1} = \frac{3}{2}$

$\therefore V_1 = \frac{6}{1 + \frac{3}{2}} = 2.4V$

$V_2 = 3.6V$

15. Ans. (A)

Sol. $v' - v = \frac{gh}{c^2} v = 1.8Hz$

16. Ans. (A)

Sol. $h\nu' = h\nu = \frac{GMh\nu}{c^2 R}$

$\Rightarrow v' = v \left(1 - \frac{GM}{c^2 R} \right)$

17. Ans. (C)

18. Ans. (C)

Sol. (Q. 17 and 18)

Least count = 0.01 mm

Error is -ve as zero is not visible.

Zero error = -1mm + 40 × 0.01 mm = -0.60 mm

Reading = (5 + 0.26) mm = 5.26 mm

Thickness = (5.26 + 0.60) mm = 5.86 mm

PART-2 : CHEMISTRY**SOLUTION****SECTION-I**1. **Ans. (D)****Sol.** (A) $E_{a_f} < E_{a_b}$

(B) Reaction is exothermic, so if we increase temperature, reaction much faster in backward direction.

(C) $\Delta H = E_{a_f} - E_{a_b} = -ve$, Exothermic(D) $\left(\frac{K}{A}\right)_f = e^{-E_f/RT}$, $\left(\frac{K}{A}\right)_b = e^{-E_b/RT}$ ($\because E_f < E_b$) $\therefore$ Fraction of activation molecules $\left(\frac{K}{A}\right)$ is higher in forward reaction2. **Ans. (D)**3. **Ans. (B)****Sol.** $HCl \xrightarrow{H_2O} H^+_{aq.} + Cl^-_{aq.}$

other species do not ionise in water.

4. **Ans. (D)****Sol.** when ppt of AgCl starts

$$K_{sp(AgCl)} = [Ag^+][Cl^-]$$

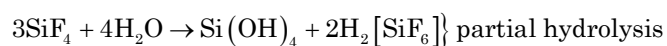
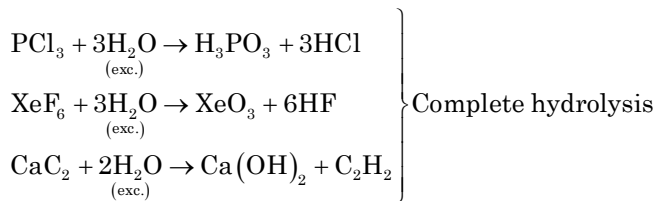
$$\therefore [Ag^+]_{left} = \frac{10^{-10}}{0.01} = 10^{-8}, \text{ which is obtained}$$

after ppt of AgBr.

$$K_{sp(AgBr)} = [Ag^+][Br^-]_{left}$$

$$[Br^-]_{left} = \frac{10^{-12}}{10^{-8}} = 10^{-4}$$

$$\% \text{ of left } [Br^-] = \frac{10^{-4}}{10^{-2}} \times 100 = 1$$

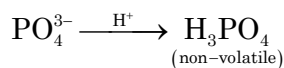
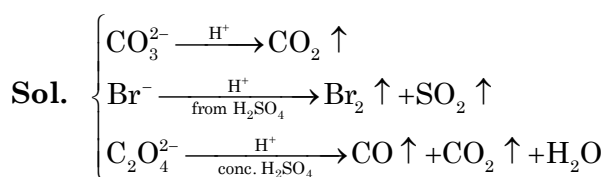
5. **Ans. (D)**6. **Ans. (C)****Sol.**7. **Ans. (D)**

$$\text{Sol. } Z = 1 - \frac{a}{RT V_m}$$

$$\text{Slope} = \frac{a}{R}$$

$$a = \frac{0.4}{1.64} \times 0.082 = 2 \times 10^{-2}$$

$$\boxed{y = 2}$$

8. **Ans. (B,C)**9. **Ans. (A, B, D)**10. **Ans. (A,C,D)**

$$\text{Sol. } W = \frac{EIt}{96500} = \frac{63.5 \times 0.15 \times 52.3 \times 60 \times 60}{96500 \times 2}$$

weight deposited = 9.292 g

$$\text{Initial total weight of Cu} = \frac{242}{1000} \times 1 \times 63.5 = 15.36 \text{ g}$$

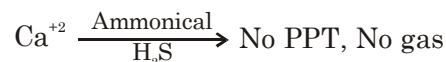
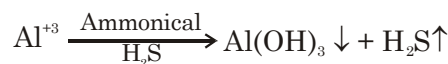
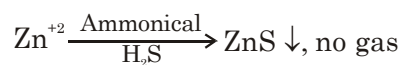
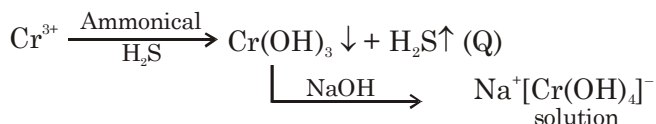
So, remaining amount of Cu in solution

$$= 15.36 - 9.292 = 6.075 \text{ g}$$

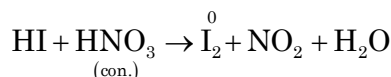
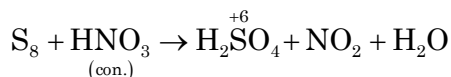
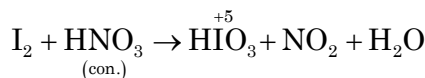
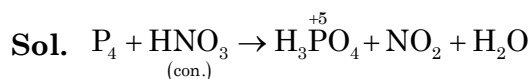
 $\therefore$ Final molarity

$$= \frac{6.075}{63.5} \times \frac{1000}{242} = 0.395 \approx 0.4$$

$$\therefore \frac{x}{10} = 0.4 \Rightarrow x = 4$$

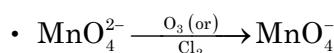
11. **Ans. (C,D)**12. **Ans. (A, C)****Sol.**

13. Ans. (A, C)



14. Ans. (A, B, C)

Sol. $\bullet$ K_2MnO_4 is stable in alkaline medium but disproportionates in acidic or neutral medium.



15. Ans. (C)

16. Ans. (B)

Sol. $M = \frac{n}{w} \times 1000 = \frac{2}{2000} \times 1000 = 1, \boxed{M=1}$

$\Delta T_f = m \times K_f \times i$

$3.84 = 1 \times 5.12 \times i \Rightarrow i = 0.75$

$0.75 = 1 + \left(\frac{1}{2} - 1\right) \alpha \Rightarrow \boxed{\alpha = 0.5}$

17. Ans. (C)

18. Ans. (D)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A)

Sol. $\sin x = 2 + [\sin x] + [-\cos x] \dots (i)$

$\{\sin x\} = 2 + [-\cos x]$

$0 \leq \{\sin x\} < 1$

and $2 + [-\cos x]$ can have values 1, 2, 3

Hence no solution

2. Ans. (C)

Sol. $\frac{\pi}{2} + \sin^{-1}(\sin 5) + \frac{\pi}{2} - \cot^{-1}(\cot 7)$

$-\sec(\sec^{-1}(-3)) + \sin\left(\sin^{-1}\frac{1}{2}\right)$

$= \frac{\pi}{2} + 5 - 2\pi + \frac{\pi}{2} - (7 - 2\pi) + 3 + \frac{1}{2}$

$= \pi + \frac{3}{2}$

3. Ans. (A)

Sol. The vertices of the triangle will be (0, 0),

$\left(\frac{1}{l+m}, \frac{1}{l+m}\right)$ and $\left(\frac{1}{l-m}, \frac{1}{l-m}\right)$.

Let the circumcentre of the triangle be (x, y)

then $x = \frac{l}{l^2 - m^2}, y = \frac{-m}{l^2 - m^2}$

$\Rightarrow 2(x^2 + y^2) = x + y$

4. Ans. (C)

Sol. Let $A\left(ct_1, \frac{c}{t_1}\right), B\left(ct_2, \frac{c}{t_2}\right), C\left(ct_3, \frac{c}{t_3}\right), D\left(ct_4, \frac{c}{t_4}\right)$

satisfy the circle $x^2 + y^2 + 2gx + 2fy + 3 = 0$

i.e. $c^2t^4 + 2gct^3 + 3t^2 + 2fct + c^2 = 0$ has

t_1, t_2, t_3, t_4 as roots(i)

Solving orthocentre of $\triangle ABC$ is get

$\left(-ct_4, -\frac{-c}{t_4}\right)$ (using $t_1t_2t_3t_4 = 1$)

Solving orthocentre of $\triangle ABD$ is $\left(-ct_3, -\frac{-c}{t_3}\right)$

Solving orthocentre of $\triangle ABC$ is $\left(-ct_2, -\frac{-c}{t_2}\right)$

The above three points satisfy

$x^2 + y^2 + mx + ny + P = 0$

$c^2t^4 - mct^3 + pt^2 - nct + c^2 = 0,$

$t_1t_2t_3t_4$ as roots(ii)

comparing (i) and (ii)

$2g = -m, 2f = -n, p = 3$

$m + n + p = -2g - 2f + 3 = 21$

5. Ans. (B)

Sol. $\alpha + \beta + \gamma = 2$ (i)

$\alpha\beta + \beta\gamma + \gamma\alpha = 3$ (ii)

$\alpha\beta\gamma = 4$ (iii)

$\frac{1}{\alpha\beta + \gamma - 1} = \frac{1}{\alpha\beta - \alpha - \beta + 1}$ (using (i))

$= \frac{1}{(\alpha - 1)(\beta - 1)}$

$\therefore \frac{1}{\alpha\beta + \gamma - 1} + \frac{1}{\beta\gamma + \alpha - 1} + \frac{1}{\alpha\gamma + \beta - 1}$

$= \frac{\alpha + \beta + \gamma - 3}{(\alpha - 1)(\beta - 1)(\gamma - 1)}$

$= \frac{-1}{\alpha\beta\gamma - (\alpha\beta + \beta\gamma + \gamma\alpha) + (\alpha + \beta + \gamma) - 1}$

$= \frac{-1}{4 - 3 + 2 - 1} = \frac{-1}{2}$

6. Ans. (C)

Sol.

$$\begin{array}{c} 8 \quad 12 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ 5 \quad 3+2 \quad 5 \quad 5 \end{array} \text{ or } \begin{array}{c} 8 \quad 12 \\ \swarrow \quad \searrow \quad \swarrow \quad \searrow \\ 4+1 \quad 4+1 \quad 5 \quad 5 \end{array}$$

$${}^8C_5 \times {}^3C_3 \times {}^{12}C_2 \times \frac{{}^{10}C_5}{2!} + \frac{{}^8C_4}{2!} \times ({}^{12}C_2 \times 2!) \times \frac{{}^{10}C_5}{2!}$$

$$= \frac{{}^{10}C_5}{2!} \times {}^{12}C_2 [{}^8C_5 + {}^8C_4] = {}^{10}C_5 \times 33 \times {}^9C_5$$

7. Ans. (D)

Sol. General term of $(\sqrt{2} + \sqrt[3]{3} + \sqrt[6]{7})^{10}$ is

$$\frac{10!}{n_1!n_2!n_3!} (\sqrt{2})^{n_1} (\sqrt[3]{3})^{n_2} (\sqrt[6]{7})^{n_3}$$

where $n_1 + n_2 + n_3 = 10, 0 \leq n_1, n_2, n_3 \leq 10 \in W$

n_1	n_2	n_3
10	0	0
4	6	0
4	0	6

sum of rotational terms is

$$2^5 + {}^{10}C_4 \times 4 \times 9 + {}^{10}C_4 \times 4 \times 7 = 13472$$

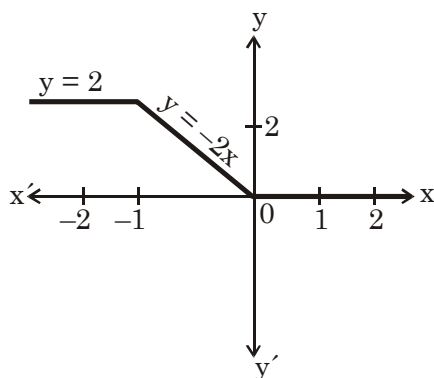
8. Ans. (A,C,D)

Sol. $f(x) = \min \phi(t), -3 \leq t \leq x$

$$\phi(x) = ||x-1| - |x+1||$$

From the graph

$f(x)$ is non-differentiable at $x = 0, -1$



$$f'(100) = 0$$

$$\int_{-3}^{10} f(x) dx = \int_{-3}^{-1} 2 dx + \int_{-1}^0 (-2x) dx + \int_0^{10} 0 dx$$

$$2[-1+3] - [x^2]_{-1}^0 = 5$$

9. Ans. (A,B,C)

Sol. $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) \Rightarrow a + b + c = 2a + b$

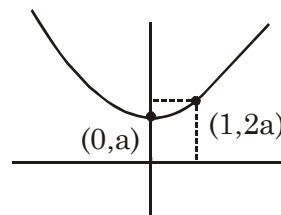
$$\Rightarrow c = a \quad \dots\dots(i)$$

$$f'(-1) = f'(1) \Rightarrow 2a + b = 2a$$

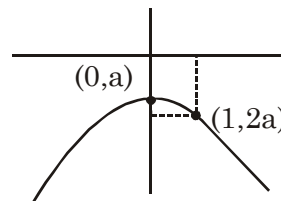
$$\Rightarrow b = 0 \quad \dots\dots(ii)$$

$$\therefore f(x) = \begin{cases} a(x^2 + 1), & x < 1 \\ 2ax, & x \geq 1 \end{cases}$$

When $a > 0$, the graph of f is



When $a < 0$, the graph of f is



Clearly (A), (B) and (C) are correct.

10. Ans. (B,C,D)

Sol. $\frac{\sin x}{x}$ is decreasing function in $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$

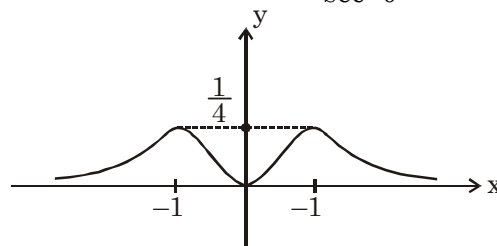
$$\therefore \frac{3\sqrt{3}}{2\pi} < \frac{\sin x}{x} < \frac{3}{\pi}$$

$$\frac{3\sqrt{3}}{2\pi} \left(\frac{\pi}{3} - \frac{\pi}{6} \right) < \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin x}{x} dx < \frac{3}{\pi} \left(\frac{\pi}{3} - \frac{\pi}{6} \right)$$

$$\frac{\sqrt{3}}{4} < \text{ques} < \frac{1}{2}$$

11. Ans. (A,B,D)

$$\text{Sol. } f(\tan \theta) = \sin^2 \theta \cos^2 \theta = \frac{\tan^2 \theta}{\sec^4 \theta}$$



Put $\tan \theta = x$

$$f(x) = \frac{x^2}{(1+x^2)^2} \quad \forall x \in \mathbb{R}$$

$$2 \times \int_0^\infty \frac{x^2}{(1+x^2)^2} dx = \int_0^\infty \underbrace{\frac{2x}{1+x^2}}_{II} dx$$

$$= \frac{-x}{(1+x^2)} \Big|_0^\infty + \int_0^\infty \frac{1}{(1+x^2)} dx$$

$$0 + \tan^{-1} x \Big|_0^\infty = \frac{\pi}{2}$$

$$a = 1, b = 2$$

12. Ans. (B,C,D)

Sol. $Z^2 = Z + |Z|^2 - \frac{2}{|Z|^3}$

Put $Z = x + iy$ and comparing real and imaginary part

$$x^2 - y^2 = x + x^2 + y^2 - \frac{2}{(x^2 + y^2)^{\frac{3}{2}}} \text{ and } 2xy = y$$

Case-I

$$y = 0$$

$$x^2 = x + x^2 - \frac{2}{x^3}$$

$$\Rightarrow x^4 = 2$$

$$\Rightarrow x = \pm 2^{\frac{1}{4}}$$

Case-II

$$x = \frac{1}{2}$$

$$\frac{1}{4} - y^2 = \frac{1}{2} + \frac{1}{4} + y^2 - \frac{2}{\left(\frac{1}{4} + y^2\right)^{\frac{3}{2}}}$$

$$\left(\frac{1}{4} + y^2\right) = \frac{1}{\left(\frac{1}{4} + y^2\right)^{\frac{3}{2}}}$$

$$\frac{1}{4} + y^2 = 1$$

$$\Rightarrow y = \pm \frac{\sqrt{3}}{2}$$

13. Ans. (B,C,D)

Sol. $T_r = \frac{4r^2 - 2}{4r^4 + 1} = \frac{4r^2 - 2}{(2r^2 + 1)^2 - (2r)^2}$

$$= \frac{2r - 1}{2r^2 - 2r + 1} - \frac{2r + 1}{2r^2 + 2r + 1}$$

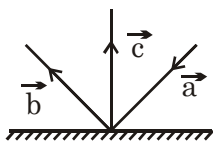
$$\sum_{r=1}^n T_r = 1 - \frac{2n + 1}{2n^2 + 2n + 1}$$

$$\Rightarrow S_n = \frac{2n + 1}{2n^2 + 2n + 1}$$

14. Ans. (A,B)

Sol. $\vec{c}$ is the angle bisector

$$\vec{c} = \frac{\vec{b} - \vec{a}}{|\vec{b} - \vec{a}|}$$



15. Ans. (D)

Sol. Let the circle be $x^2 + y^2 = a^2$ and let the centres of a rectangular hyperbola be (h, k) . Let the given points on the circle be $(a \cos \theta_i, a \sin \theta_i)$, $i = 1, 2, 3, \dots$

$$\Rightarrow \frac{a \cos \theta_1 + a \cos \theta_2 + a \cos \theta_3 + a \cos \theta_4}{4}$$

$$= \frac{0 + h}{2} = \frac{h}{2}$$

$$\Rightarrow \sum_{i=1}^5 a \sin \theta_i - a \cos \theta_5 = 2h$$

Since the points are given, L and M are known. Therefore, we have

$$a \cos \theta_5 = L - 2h, a \sin \theta_5 = M - 2k$$

$$a^2 = (2h - L)^2 + (2k - M)^2$$

Hence (h, k) lies on the curve

$$\left(x - \frac{L}{2}\right)^2 + \left(y - \frac{M}{2}\right)^2 = \left(\frac{a}{2}\right)^2$$

which is a circle of radius $\frac{a}{2}$

16. Ans. (A)

Sol. Let A, B, C, D have coordinate (x_i, y_i) $i = 1, 2, 3, 4, \dots$ then

$$\frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{3h}{2}, \dots (i)$$

$$\frac{y_1 + y_2 + y_3 + y_4}{4} = \frac{3k}{2}, \dots (ii)$$

Let the centroid of ΔABC be (α, β) , then

$$\alpha = \frac{x_1 + x_2 + x_3}{3}, \beta = \frac{y_1 + y_2 + y_3}{3}$$

From equation (i) and (ii) we get,

$$3\alpha + x_4 = 6h \text{ and } 3\beta + y_4 = 6k$$

But (x_4, y_4) lies on $x^2 - y^2 = 9a^2$

$$(6h - 3\alpha)^2 - (6k - 3\beta)^2 = 9a^2$$

$$(\alpha - 2h)^2 - (\beta - 2k)^2 = a^2$$

$\therefore$ Locus of (α, β) is $(x - 2h)^2 - (y - 2k)^2 = a^2$

17. Ans. (C)

Sol. D.R of normal vector to plane is $\langle 1 + \gamma, 2 - \gamma, 1 - \gamma \rangle$ and D.R of line parallel to the plane is $\langle 3, -1, 2 \rangle$

Since both are perpendicular

$$\therefore 3(1 + \gamma) - 1(2 - \gamma) + 2(1 - \gamma) = 0$$

$$\Rightarrow 3 + 3\gamma - 2 + \gamma + 2 - 2\gamma = 0$$

$$\gamma = -\frac{3}{2}$$

18. Ans. (D)

Sol. Plane $x + 2y + z = 1$ will be perpendicular to plane $(1 + \gamma)x + (2 - \gamma)y + (1 - \gamma)z - (1 + 2\gamma) = 0$

$$\therefore 1(1 + \gamma) + 2(2 - \gamma) + 1(1 - \gamma) = 0$$

$$\gamma = \frac{5}{2}$$